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The Reproduction of Academic Status Hierarchies

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Abstract:
American academia is steeply hierarchical: a small set of colleges and universities produce a majority of faculty, and upward mobility is rare. Existing work documents this pattern, but the causes of its emergence and persistence are less clear. Here I show that steep hierarchies are easily sustained by structural inequality in the production of candidates combined with weak status signals. A matching model demonstrates the mechanism. When positions are scarce relative to candidates, a small preference for prestige is amplified into highly concentrated outcomes, a condition common to academic and other elite labor markets. To measure both, I link IPEDS bachelor's records and the NCSES Survey of Earned Doctorates to a census of tenure-track faculty at PhD-granting US universities, placing undergraduate production, doctorate production, and faculty placement on a common institution-by-field grid. The distribution of undergraduate production alone reproduces a median 81% of observed hierarchy, and 94% once selection into the faculty pool is included. Calibrated to the remainder field by field, the model reproduces observed hierarchy at a median status weight of 0.016 (range 0.001–0.240) — a preference that tips the choice between a candidate from the top-ranked institution and one from the bottom-ranked by about 1.6 percentage points. Status hierarchies can therefore persist without strong or widely shared agreement on the status order; small behavioral advantages, operating on top-heavy supply distributions and tight labor markets, are enough to sustain stark macro-level inequalities.

Elite labor markets are often dominated by graduates of a small number of prestigious institutions. The eight Ivy League universities educate less than 1% of undergraduates in the United States but account for over 20% of journalists at the New York Times and Wall Street Journal, more than 25% of Fortune 500 CEOs, and over 35% of presidents and vice presidents (Wai et al. 2024). Where the route to the job runs through advanced training, concentration is even sharper. Harvard and Yale alone produce 45% of Supreme Court clerks (George, Yoon, and Gulati 2023), and roughly 20% of doctoral institutions produce 80% of American faculty (Clauset, Arbesman, and Larremore 2015; Wapman et al. 2022). This pattern is not limited to the United States. Elite careers in other countries like England, France, South Korea, or Japan run through similarly small sets of prestigious schools (Holt-White et al. 2025; Dudouet and Joly 2010; Han 2024; Cutts 2015; Ivanovska Hadjievska and Arvidson 2026). I refer to the concentration of graduates from a small set of universities in elite careers as *institutional status sorting*. The recurrence of institutional status sorting across national contexts and occupations demonstrates the importance of this organizational form in modern status systems.

Institutional status sorting constitutes one of the last steps in the process of elite reproduction. Sociologists have described how elites accrue advantages beginning in the family, primary and secondary schooling, continuing through college and professional or graduate training, and culminating in access to privileged positions in science, politics, law, business, or culture (Lareau 2003; Khan 2011; Rivera 2016; Useem and Karabel 1986; Cookson and Persell 2008; Persell and Cookson 1985). Institutional status sorting therefore comes at the end of a long process, but it is key in legitimizing the status order of a society (Bourdieu 1998; Bowles and Gintis 2011). Prestigious educational institutions control access to elite careers while maintaining the perception of a meritocracy.

One striking feature of these hierarchies is their persistence. Despite substantial changes in the social composition of elites, in the fields they dominate, and in the routes into privileged positions (Karabel 2005; Benveniste 2025), the pipelines connecting prestigious schools to prestigious positions have changed remarkably little. The Ivy League, the Grandes Écoles, and Oxbridge are all examples of it (Reeves and Friedman 2024; Dudouet and Joly 2010). Here I show how weak status signals and common structural conditions of prestigious labor markets can easily sustain the levels of institutional status sorting we observe in many careers, making the perpetuation of these hierarchies unsurprising.

Existing explanations typically locate the reproduction of institutional sta-

tus hierarchies in the advantages attached to elite education. Symbolic accounts emphasize how prestigious institutions confer cultural, social, or symbolic capital, making their graduates more valuable in elite labor markets (Bourdieu 1998; Collins 1979). These advantages may operate through elite networks, cultural matching, the cultivation of elite career aspirations, or even more simply, through explicit commitments to hiring from certain prestigious institutions (Rivera 2012, 2011; Fogel, Hoopes, and Liu 2023; George, Yoon, and Gulati 2023; Binder, Davis, and Bloom 2016). Human-capital accounts instead emphasize that institutional prestige may correlate with student quality, training, productivity, or other attributes valued by employers (Becker 1994; Way et al. 2019; Zhang et al. 2022). Much of the empirical work to date focuses on disentangling the two (Headworth and Freese 2016; Way et al. 2019; Zhang et al. 2022; Dale and Krueger 2002). Here instead I treat institutional prestige as a status signal (Podolny 2005) — one that may carry information about a candidate's quality or merely the symbolic weight of their origin. What matters for the model I develop is how status is perceived by the actors in a labor market and how much weight they ascribe to it. That is, the strength of status as a signal but not its content. Neither symbolic nor human-capital accounts establish how strong that signal must be to produce the concentration observed in elite labor markets.

I show that the advantages required to sustain these hierarchies are much smaller than their degree of concentration suggests. In many elite labor markets, candidates are not produced evenly across institutions. Some institutions educate more students, send more students to graduate school, and produce more candidates for elite positions than others. These differences compound across the pipeline before final hiring or selection occurs. At the same time, elite labor markets are often highly unbalanced, with many qualified candidates competing for relatively few positions. Under these structural conditions, a weak status signal can be amplified into highly concentrated aggregate outcomes. This means that status signals, in the form of symbolic closure or human-capital advantages, might play a much smaller role in the reproduction of elite overrepresentation than previously thought once structural factors like unequal production and labor oversupply are taken into account.

I take the American academic labor market as an example of institutional status sorting. The overrepresentation of elite universities in faculty origins has been consistently observed across disciplines (Wapman et al. 2022; Clauset, Arbesman, and Larremore 2015; Markovic 2020; Grunspan et al. 2024), and faculty are also drawn disproportionately from high-income and highly educated families (Morgan et al. 2022; Abramitzky et al. 2024; Stansbury and Schultz 2023; Spoon 2025). However, the causes of this institutional

concentration and its persistence are less clear. First, I use a stylized simulation model to show how weak status signals can generate highly concentrated outcomes under common structural conditions. In the model, candidates for academic positions and hiring departments rank one another using a weighted combination of institutional prestige and random noise; varying the weight on prestige shows how small behavioral advantages are amplified by the oversupply of labor and production inequality characteristic of elite careers. I use this simple model to isolate the core mechanism, and develop a more general version in the Appendix that allows candidate quality and selection out of academic careers to vary independently (Appendix A.3). Second, I link undergraduate degree records, doctorate records, and a census of US faculty into a single institution-by-field panel, making the academic pipeline observable from undergraduate production through faculty placement. I use it to ask how much of the hierarchy in faculty hiring is expected given the unequal supply of candidates at each stage, and how much remains after selection into academic careers. I then calibrate the matching model to the remainder, recovering estimates for each field of the strength of status signal that reproduces its observed hierarchy.

Results

A Matching Model of Status Sorting

I use a stylized matching simulation to show how weak status signals can produce steep status hierarchies. One side of the market consists of candidates, such as PhD recipients seeking faculty positions; the other consists of institutions with vacancies, such as departments hiring assistant professors. The key structural parameter is the supply-to-demand ratio, r, defined as the number of candidates per available position. When $r = 1$, the market is balanced. When $r > 1$, positions are scarce relative to the supply of candidates.

Candidates and departments rank one another using a weighted combination of institutional prestige and idiosyncratic noise:

$$U_{ij}^{C} = \alpha P_j + (1 - \alpha)\epsilon_{ij}^{C},$$
$$U_{ji}^{D} = \alpha P_i + (1 - \alpha)\epsilon_{ji}^{D}.$$

Candidate i evaluates department j according to U_{ij}^{C}, while department j evaluates candidate i according to U_{ji}^{D}. Here, P_j is the normalized status rank of department j, P_i is the status of candidate i's origin institution, and $\alpha \in [0, 1]$ controls the weight placed on prestige. The error terms ϵ are independent

draws from a uniform distribution on the unit interval. The uniform is the maximum-entropy distribution on $[0, 1]$, so this specification assumes only that idiosyncratic judgment is bounded on the same scale as prestige, and nothing further about how evaluators' idiosyncratic assessments are distributed within those bounds. Because the hierarchy a given α produces can depend on the shape of that distribution and not only on its spread, the Appendix repeats the analysis across a set of bounded distributions — Beta(a, a) on $[0, 1]$ for $a = 1, 2, 5, 7$, of which the uniform is the $a = 1$ case (Fig. A2). Amplification by oversupply holds across all four. The distributions differ only where the status signal is weakest, with the uniform the most permissive case, and the curves converge as α rises. When $\alpha = 0$, rankings are fully random; when $\alpha = 1$, rankings are based solely on institutional status.

I match candidates to vacancies using the Gale–Shapley algorithm (Gale and Shapley 1962). The algorithm lets candidates propose in order of their preferences while institutions retain their most-preferred candidates up to capacity, producing a stable allocation with no unmatched candidate–institution pair that would prefer one another to their assigned matches. When $r = 1$, all candidates are matched to a position. When $r > 1$, some candidates are not hired, simulating the selection of candidates out of the academic labor market. The model assumes that the only signal available to actors in the market is institutional prestige. This simplified version of the academic labor market serves to illustrate how individual preferences can be amplified under certain structural conditions. Alternative matching procedures, including department-proposing, decentralized, and shortlist-based matching, yield similar results where the status signal is weak (Appendix A.2). A more general version of the model, in which departments also reward candidate quality unrelated to origin status and candidates select out of academic careers before matching, reproduces observed hierarchies with a weak status signal when quality carries little weight or selection favors candidates from high-status origins (Appendix A.3).

Following Wapman et al. (2022), I measure the degree of hierarchical organization using the share of anti-hierarchical moves. In an idealized status hierarchy, candidates move only from higher-ranked to lower-ranked institutions. A move from a lower-ranked origin to a higher-ranked destination is therefore anti-hierarchical. Lower rates of anti-hierarchical mobility indicate a steeper hierarchy and a higher concentration of positions in the hands of candidates from more prestigious institutions.

I simulate the matching market under three supply distributions. In the uniform distribution, all institutions produce the same number of candidates. In the top-heavy distribution, higher-ranked institutions produce more can-

didates; in the bottom-heavy distribution, lower-ranked institutions produce more candidates. For the simulations in Figure 1, candidate supply is proportional to a beta density evaluated over institutions' normalized status ranks, using shape parameters (4, 1.5) for the top-heavy distribution and (1.5, 4) for the bottom-heavy distribution (Appendix Fig. A1). The resulting counts are normalized to the desired total supply. For each parameter combination, the simulation matches between 250 (at $r = 1$) and 1,250 candidates (at $r = 5$) to 50 departments with five vacancies each and averages the results over 100 independent iterations. Varying both the supply distribution and the supply-to-demand ratio allows the analysis to identify when weak status signals are sufficient to generate highly concentrated outcomes.

Figure 1 shows how the degree of hierarchy generated by the matching process depends jointly on status sensitivity, the supply-to-demand ratio, and the distribution of candidate origins. When $\alpha = 0$ (status indifference), the three supply distributions already produce markedly different outcomes. The uniform distribution yields a hierarchy strength of approximately 50%, whereas the top-heavy and bottom-heavy distributions yield values of roughly 73% and 27%, respectively. Thus, the distribution of candidate origins alone can generate either concentrated or anti-hierarchical mobility in the absence of status-sensitive matching.

Under a uniform supply distribution (panels A and D), increasing α has little effect when supply matches demand ($r = 1$); hierarchy strength remains close to 50%. When candidates outnumber positions, however, status sensitivity produces progressively more concentrated outcomes. The relationship is approximately linear when the market is only slightly unbalanced ($r = 1.2$), but at higher ratios hierarchy strength rises steeply at low values of α and then levels off. At $r = 5$, hierarchy strength rises sharply even at low values of α, reaching approximately 86% when $\alpha = 0.1$. This corresponds to fewer than 14% of hiring moves being upward in a market where prestige barely tips any individual comparison: at $\alpha = 0.1$, an evaluator on either side of the market ranks the higher-status of two alternatives above the other 55% of the time when the two are separated by half the status range, and 60% of the time when one is the top-ranked institution and the other the bottom-ranked. I refer to weights in this range, $\alpha \leq 0.1$ (shaded in Fig. 1), as weak status signals. The same status sensitivity therefore produces substantially different aggregate outcomes depending on the degree of market imbalance.

The top-heavy distribution (panels B and E) produces a similar pattern. Because a larger share of candidates comes from high-status institutions, the baseline level of hierarchical organization is already strong. Increasing status sensitivity reinforces hierarchy strength across supply-to-demand ratios,

reaching approximately 90% or more at moderate levels of status sensitivity. Here, the distribution of candidate origins and status-sensitive matching operate in the same direction: high-status candidates are more numerous and are also more likely to be selected into high-status positions.

The bottom-heavy distribution (panels C and F) produces the opposite pattern in balanced markets. At $r = 1$, hierarchy strength begins at approximately 27% and declines toward zero as α increases. At $r = 1.2$, it follows a similar, though less pronounced, pattern. These results reflect the two-sided structure of the model. Most candidates originate in lower-status institutions, while candidates prefer higher-status destinations as α increases. When nearly every candidate can be placed, this preference generates additional upward moves. Departmental preferences for candidates from high-status origins are not sufficient to offset this effect because relatively few such candidates are available. At higher supply-to-demand ratios ($r > 2$), the bottom-heavy distribution follows the broader pattern observed under the other distributions. Hierarchy strength rises sharply at low status sensitivity and then levels off.

Taken together, the simulations show that weak status signals can generate a high degree of hierarchical organization under particular structural conditions. The effect is strongest when supply is top-heavy or when a uniform supply distribution is combined with substantial excess supply. It is weakest, and may even be reversed, when supply is bottom-heavy and the market is balanced. Aggregate institutional sorting therefore cannot be inferred from the strength of status sensitivity alone. It also depends on who enters the market and the amount of competition for each position.

Linking the Academic Pipeline

The simulations show how weak status signals generate steep hierarchies when candidates outnumber positions. This is common in American academia, where doctoral programs award far more degrees than there are faculty positions to fill (Larson, Ghaffarzadegan, and Xue 2014; Alberts et al. 2014; Cyranoski et al. 2011; Sarrico 2022). I follow the academic pipeline to observe the process of hierarchy formation by linking three sources — IPEDS bachelor's-degree records, the NCSES Survey of Earned Doctorates, and the census of US faculty assembled by Wapman et al. (2022) — which together place undergraduate production, doctorate production, and faculty placement on a common institution-by-field grid spanning 31 scientific fields and 325 institutions. I rank institutions using the SpringRank algorithm (De Bacco, Larremore, and Moore 2018), the network-based method used by Wapman et al. (2022), under which an institution ranks above another when its graduates

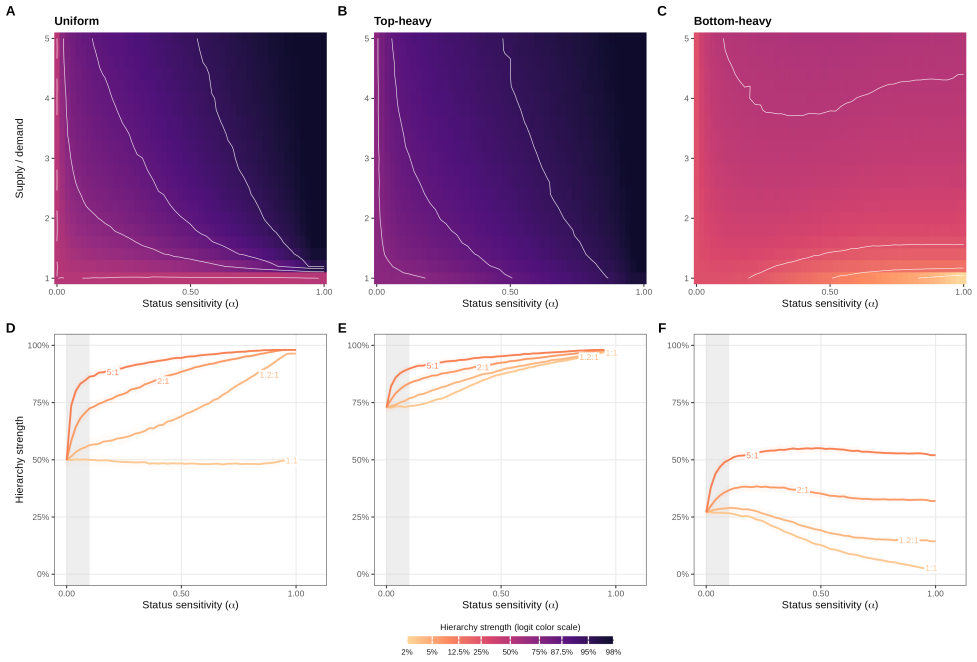


Figure 1: Weak status signals can generate strong status hierarchies. Heat maps show mean hierarchy strength across status sensitivity, α, and the supply-to-demand ratio, r, for uniform (A), top-heavy (B), and bottom-heavy (C) candidate-origin distributions. Line plots show the corresponding results at selected values of r: 1, 1.2, 2, and 5 candidates per position (D–F). Hierarchy strength is defined as one minus the share of up-hierarchy moves. Ribbons indicate the 10th–90th percentile range across simulation replicates. The shaded region marks weak status signals, $\alpha \leq 0.1$, under which an evaluator prefers a candidate or department of the highest status over one of the lowest status at most 60% of the time. The same color scale is used across the heat maps. Because the three distributions occupy opposite ends of the range, color is mapped on a logit scale, so equal color steps correspond to equal steps in the log-odds of a down-hierarchy move; white lines are contours drawn at the color-bar tick values.

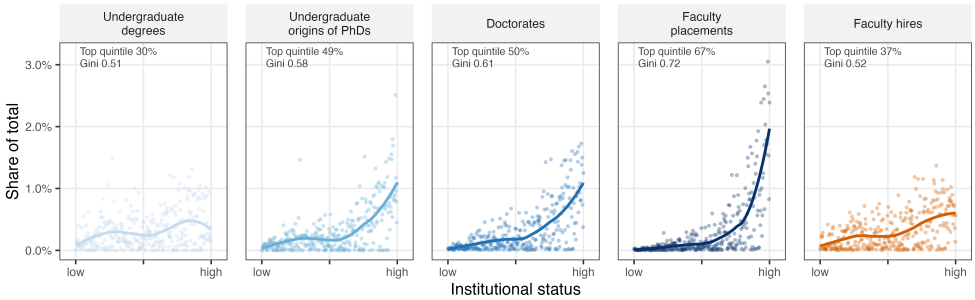


Figure 2: Candidate production concentrates among high-status institutions across the academic pipeline; faculty positions do not. Each point is one of the 325 institutions with an Academia-level SpringRank, plotting its share of that stage's total against its status rank, from lowest to highest. Lines are locally weighted regressions. Labels report the share of the stage held by the highest-status quintile and the Gini coefficient. The undergraduate panel covers only bachelor's degrees awarded by these doctoral institutions and so excludes liberal arts colleges. Restricting to ranked institutions understates concentration: across all 460 SED doctorate-producing institutions the Gini for doctorates is 0.67, against 0.61 here.

are hired by that institution more often than the reverse. I use this dataset to ask how much of the status hierarchy in faculty hiring is already present in the distribution of candidates, how much remains once I account for selection and sorting, and how strong a status signal is needed to generate the remainder.

Elite Dominance across the Pipeline

Elite institutions account for a larger share of the candidate pool at each successive stage of the academic pipeline (Fig. 2). Institutions in the highest-status quintile award 30% of undergraduate degrees, educate 49% of those who go on to earn doctorates, produce 50% of doctorates, and train 67% of faculty who are eventually placed. Yet they hold only 37% of faculty positions.

That gap is what links hierarchy to concentration. Elite institutions supply 67% of placed faculty but employ only 37%, so at least 30 of those 67 percentage points must be placed below the top quintile. Downward flow — hierarchy — is therefore generated by the shape of the candidate pool before any status-sensitive behavior enters the market. Some of what looks like status-based sorting is therefore arithmetic.

I use four null models to establish how much of the observed hierarchy can be accounted for by inequality in candidate production (Fig. 3). Each pairs a different candidate-origin distribution with the same empirical distribution of faculty positions. Null 1 draws candidate origins in proportion to field-specific undergraduate production; Null 2 uses the baccalaureate institutions of doctorate recipients; Null 3 uses doctorate production by doctoral institution; Null 4 uses the observed distribution of faculty placements. Because the nulls differ only in the supply margin, differences between adjacent nulls give an accounting decomposition of the hierarchy, without identifying causal effects or individual transitions.

Null 1 reproduces a median 81% of the observed hierarchy across fields (median hierarchy strength of 72% under Null 1, compared with 89% observed). Unequal undergraduate production and the observed distribution of faculty positions thus generate substantial downward mobility even when candidates are matched to positions without regard to status.

Null 2 incorporates the distribution of baccalaureate origins among individuals who eventually receive doctorates. Relative to Null 1, this raises hierarchy strength by a median 0.4 percentage points, but the contribution is negative in 14 of the 31 fields. In those, the doctorate population is drawn disproportionately from lower-status undergraduate institutions, so selection into and through doctoral study works against hierarchy rather than reinforcing it.

Null 3 replaces undergraduate origins with doctoral institutions, combining mobility between undergraduate and doctoral institutions with doctoral admissions and completion. This raises hierarchy strength by a median 2.1 percentage points and is negative in 13 of the 31 fields, so the shift from baccalaureate to doctoral origins adds little to the hierarchy already implied by undergraduate production.

Null 4 uses the observed distribution of faculty placements by doctoral institution while preserving the empirical faculty-demand structure. The difference between Nulls 3 and 4 captures the transition from doctoral production into the faculty candidate pool, including selection into academic careers. This is the largest post-undergraduate contribution, raising hierarchy strength by a median 9.9 percentage points, and it is positive in every field.

Across fields, Nulls 1 through 4 reproduce median shares of 81%, 82%, 83%, and 94% of the observed hierarchy, respectively (Fig. 3). The median residual above Null 4 is approximately 5 percentage points, or 6% of observed hierarchy. These medians conceal meaningful sign reversals, since several fields contain stages that offset rather than reinforce hierarchy. Most of the observed hierarchy is nonetheless already present in the supply distributions

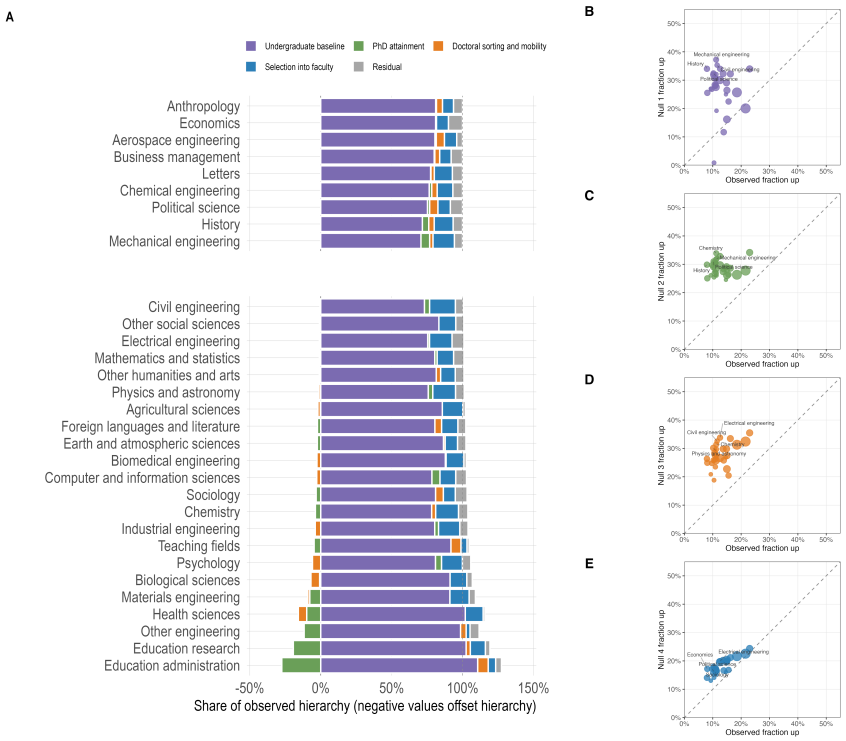


Figure 3: Decomposing observed hierarchy into mechanical and residual components. Panel A separates the undergraduate-production baseline (Null 1), PhD attainment (Null 2 minus Null 1), doctoral sorting and mobility (Null 3 minus Null 2), selection into the observed faculty population (Null 4 minus Null 3), and the residual above Null 4. Fields are grouped by whether every stage reinforces hierarchy (top) or at least one stage offsets it (bottom). Panels B–E compare the observed fraction of up-hierarchy hires with Nulls 1 through 4. Points above the dashed line indicate that the null generates more up-hierarchy hiring than observed and therefore a less steep hierarchy.

that precede faculty hiring.

Calibrating the Strength of Status Preferences

The decomposition establishes the expected contribution of supply differences to the levels of hierarchy in academic hiring. However, it does not determine

how much status sensitivity is required to account for the remaining hierarchy. The simulation model provides a way to quantify this residual. For each field, the empirical demand distribution is held fixed while the model is calibrated to the observed fraction of up-hierarchy moves. The resulting $\alpha_{\min}$ is the minimum status sensitivity required for the simulated matching process to reproduce the observed rate. Institutional status is ranked separately within each field, and the same fixed observed ranking is used for preferences and for scoring observed and simulated placements.

Across all 31 fields, the median $\alpha_{\min}$ is 0.016 (range 0.001–0.240), and 29 of the 31 fall below 0.05 (Fig. 4A), well inside the range of weak status signals. A weight of 0.016 is close to indifference. A department comparing two candidates whose origin institutions sit at opposite ends of the status ranking prefers the higher-status one 52% of the time, and 51% of the time when the two origins are separated by half the ranking. That is enough to reproduce hierarchies in which a median of only 11% of hires move up the status ranking. These estimates assume uniform idiosyncratic noise, the most permissive of the bounded distributions considered in Appendix A.1; more concentrated noise requires larger status weights to produce the same hierarchy.

Fields differ widely in how crowded their markets are (Fig. 4B). The median field produces roughly six doctorates for every new faculty position, but the range runs from fewer than two in other engineering and business management to more than thirty in the biological sciences. Twenty-three of the 31 fields produce more than five doctorates per position, placing them in the region of the simulation where small changes in the status signal generate large changes in hierarchy.

The required signal falls as markets grow more crowded. Fields with more doctorates per position reproduce their observed hierarchy at lower $\alpha_{\min}$, which is the pattern the simulations predict: where positions are scarce, departments can act on a faint preference many times over. The exceptions are the two least crowded markets in the sample, other engineering and business management, at fewer than two doctorates per position; both require signals above 0.2.

These estimates support the mechanism the simulations identify. Once the unequal production of candidates and the scarcity of positions are taken into account, the hierarchy observed in most fields is compatible with a status signal weak enough that it would be hard to detect in any single hiring decision.

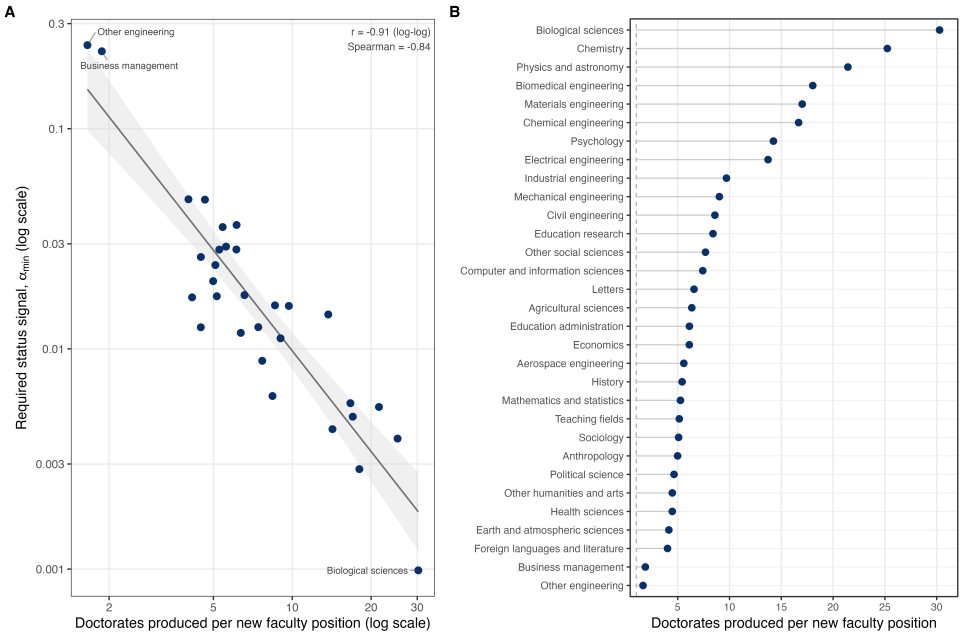


Figure 4: Fields with more crowded labor markets require weaker status signals. (A) The minimum status signal $\alpha_{\min}$ — the smallest weight on institutional prestige at which the matching model reproduces a field's observed rate of up-hierarchy hiring — against its supply-to-demand ratio. Both axes are logarithmic; the line is an ordinary least squares fit on logged values with a 95% confidence band. (B) The same ratios by field on a natural scale; the dashed line marks a balanced market of one doctorate per position. Supply is SED doctorate production for 2011–2020 and demand is Wapman new-faculty counts; points are all 31 canonical fields, each simulated with 1,000 vacancies (Methods).

Discussion

Steep status hierarchies in US academia can be sustained by a combination of structural conditions and weak status signals. Most of the hierarchy in faculty hiring is already present in the distribution of candidates before hiring begins. The institutions that produce most undergraduates produce most of the future doctoral pool, and the institutions that produce most doctorates train an even higher share of the future faculty. Once that supply structure and the scarcity of positions are taken into account, reproducing the observed hierarchy requires a status signal with a median weight of 0.016 — one that tips the choice between a candidate from the top-ranked institution and one from the bottom-ranked by about 1.6 percentage points. The mechanism belongs to a family of models in which weak individual effects produce strong aggregate outcomes (Schelling 1971; Rosen 1981; Xie, Cheng, and Zhou 2015; Acerbi and De Courson 2025; Gould 2002; Mark 1998). Hierarchies this cheap to sustain should be expected to persist, and to persist through changes in who the elites are.

This mechanism suggests a reason why academic status hierarchies are so persistent. Status is relational, meaning that an institution's standing depends partly on its associations with other institutions (Podolny 2005). By supplying a disproportionate share of entrants to elite careers, elite institutions may therefore help sustain their own status. Theories of social closure emphasize how privileged groups restrict access to desirable positions to reproduce their advantages (Bourdieu 1998). The mechanism identified here shows that structural conditions can support this reproduction even when individual advantages are small. When candidates greatly outnumber positions, modest status preferences can reinforce the dominance of institutions that already produce disproportionate shares of candidates, favoring continuity in which institutions occupy the top of the hierarchy. This perspective also recasts candidate oversupply. Although often described as a dysfunction of academic labor markets (Larson, Ghaffarzadegan, and Xue 2014; Alberts et al. 2014), oversupply may nevertheless serve the interests of the institutions that dominate them.

The same mechanism may operate beyond academia, wherever entry to coveted careers passes through long educational pipelines and candidates outnumber available positions. In such markets, institutional dominance can persist without strong or widely shared agreement about which institutions deserve their standing. Understanding elite reproduction therefore requires attention to the structural conditions of these markets, alongside the micro-level processes through which candidates are evaluated and selected.

Materials and Methods

Data Sources and Linkage

I construct a panel of US academic production by linking the NCSES Survey of Earned Doctorates (SED) and IPEDS bachelor's-degree records to the faculty hiring network of Wapman et al. (2022). The SED provides doctorate counts by institution, field and year for 2011–2020, a total of 536,661 doctorates across 460 doctorate-producing institutions and 62 fields. The hiring network records, for each subfield, the number of faculty employed at each institution by the institution that granted their doctorate. I manually map SED institutions to institutions in the hiring network. Because the two sources classify disciplines at different levels of detail, I assign each of the hiring network's 107 subfields to a broad SED field, producing 31 fields that appear in both sources (Appendix B.2). The resulting panel observes undergraduate production, doctorate production and faculty placement on a common institution-by-field grid.

I constructed the SED panel using Survey of Earned Doctorates data downloaded from the NCSES Table Builder on May 19, 2026. I retained institution records with IPEDS identifiers, restricted the data to 2011–2020, and reshaped the 62 Trend Major Field columns into doctorate counts by institution, field, and year.

The hiring network is the public `us-faculty-hiring-networks` release accompanying Wapman et al. (2022). It covers all tenure-track faculty at PhD-granting US universities. I use its subfield-level edge list, which gives the number of faculty trained at each institution and employed at each other institution, its counts of new faculty, and its Academia-level institutional prestige ranking. I matched SED institutions to hiring-network institutions by name after standardizing abbreviations, state codes, and filler words, and manually reviewed all matches.

Measuring hierarchy

Following Wapman et al. (2022), I measure hierarchy as one minus the share of hires that move up the prestige ranking, from a lower-ranked doctoral institution to a higher-ranked employer. Self-hires count as non-upward moves. For each field, observed hierarchy is computed by estimating SpringRank (De Bacco, Larremore, and Moore 2018) on the field's hiring network, restricted to institutions matched to the SED and to a hiring-closed support, in which every institution represented as a doctoral origin also employs faculty. Institutions that only employ faculty can remain.

In the stylized simulations and Appendices A.1–A.2, departments rank candidates from their own institution last, and self-hires are excluded from the denominator when computing hierarchy. Appendix A.3 allows self-hires during matching without penalizing them in departmental preferences, but likewise excludes them when computing hierarchy.

Null models

I construct four null models to quantify the hierarchy expected from unequal candidate production under random assignment to faculty positions. All models use the same 31 fields and the same analytical sample of institutions described above, preserving the observed number of faculty positions at each institution.

Nulls 1–3 draw candidate origins independently, with replacement, and randomly assign them to the observed positions. Sampling probabilities are proportional to field-specific undergraduate degree production in IPEDS during 2011–2020 (Null 1), the number of doctorate recipients by baccalaureate institution in NCSES data (Null 2), or doctorate production by doctoral institution in the SED during 2011–2020 (Null 3). Null 2 therefore conditions on eventual doctorate completion. Null 4 repeatedly exchanges employers between pairs of faculty placements, preserving the number of faculty each institution produces and employs, following Wapman et al. (2022). Self-hires are held fixed, and exchanges that would create new self-hires are rejected.

For each field, I generate 100 realizations of each null and reestimate SpringRank on every resulting network. Hierarchy is one minus the fraction of hires moving from a lower-ranked origin to a higher-ranked employer, with self-hires counted as non-upward moves. I report mean hierarchy across realizations. Differences between successive nulls provide an accounting decomposition associated with changes in the candidate-origin distribution.

Simulation and calibration

For each field, I simulate matching between doctorate recipients and hiring departments using the candidate-proposing Gale–Shapley algorithm and the utility functions defined in the main text. Institutional status is measured by the ordinal SpringRank estimated on each field's observed, SED-matched, hiring-closed network and rescaled to $[0, 1]$, with tied scores retaining tied ranks. I hold this ranking fixed for both sides' preferences and for scoring observed and simulated upward hires; I do not reestimate rankings on simulated matching outcomes. Candidate supply, positions, and observed hires are

restricted to the same ranked institutions. Idiosyncratic evaluations are independent draws from a uniform distribution on $[0, 1]$. Candidate supply follows each retained institution's SED doctorate production during 2011–2020. New-faculty counts in the hiring network measure demand, defined as faculty who entered the 2011–2020 sample within three years of receiving their doctorate. I use field-level counts when they cover at least 70% of the field's observed hires, adjusting partially covered totals by their hire coverage; otherwise, I allocate broader-domain counts across fields in proportion to observed hires. I retain the resulting demand assigned to institutions in the ranked support and redistribute that total across them in proportion to hires in the closed observed network. I simulate 1,000 vacancies in each field and scale candidate counts to preserve the supply-to-demand ratio. Institutional counts on both sides are assigned by largest-remainder rounding. The main estimates and the participation analysis in Appendix B.3 use the same rankings, institutional demand allocation, market size, and calibration routine; full participation is the same fitted scenario in both analyses.

I calibrate the status weight, $\alpha_{\min}$, to reproduce each field's observed share of upward hires, counting self-hires as non-upward moves. A coarse first-crossing search over $\alpha \in [0, 1]$ averages 12 simulated markets per grid point. I refine the crossing using 40 markets per evaluation and five bisection steps, then interpolate within the final bracket. Common random numbers are used across weights within each search phase. I evaluate the selected weight using 100 additional markets with independent seeds. If the absolute validation error exceeds both 0.002 and three Monte Carlo standard errors, I repeat the calibration with fresh seeds, doubling refinement and validation draws and adding two bisection steps per retry, for up to three retries. Across fields, the median absolute validation error is 0.0006 and the maximum is 0.0023; the median Monte Carlo standard error is 0.0008. The calibrated weight measures the status signal sufficient to approximate observed hierarchy under the model's assumptions.

Appendix

This Appendix examines the two components of the analysis in turn. Section A concerns the simulation model. The main text makes three simplifying assumptions: idiosyncratic judgment is uniformly distributed, matches are formed via Gale–Shapley, and institutional status is the only signal available to both sides of the market. Here I relax each of these assumptions. I first repeat the simulation across a family of bounded noise distributions (A.1) and

matching procedures (A.2). I then add a quality signal that departments observe independently of origin status, together with selection out of academic careers before matching (A.3). Section B of the appendix concerns the empirical calibration. I assess whether a single 2011–2020 supply window is a reasonable snapshot of doctorate production, given that the faculty data record a stock of past hires (B.1), and whether the estimates are sensitive to how fields are harmonized across data sources (B.2). Last, I separate selection into faculty careers from sorting among those selected, and recalibrate the sorting weight conditional on observed faculty origins (B.3).

A. Extensions and robustness of the simulation model

Figure A1 shows the candidate-origin distributions used in the baseline simulation. Each institution has the same number of vacancies, while its share of candidate supply varies with origin status. Changing the supply-to-demand ratio scales the total number of candidates while preserving the relative supply weights, subject to rounding to whole candidates.

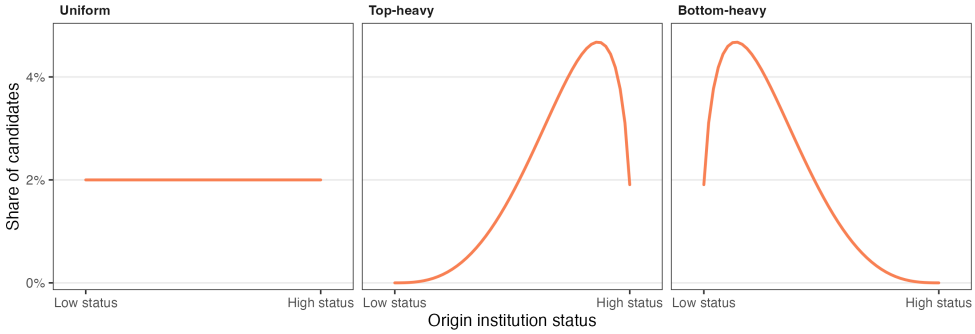


Figure A1: Candidate supply distributions in the matching model. Lines connect the supply weights for 50 institutions ordered from low to high status. Uniform supply assigns each institution 2% of candidates. Top-heavy and bottom-heavy supply weights are proportional to beta densities with shape parameters (4, 1.5) and (1.5, 4), respectively, evaluated at status midpoints and normalized to sum to one. All three cases have the same total supply at a given supply-to-demand ratio.

A.1 Robustness to distinct noise distributions The matching model in the main text draws the idiosyncratic term ϵ from a uniform distribution on $[0, 1]$.

Keeping the noise on the same bounded scale as prestige lets α be read as the share of a fixed evaluation budget given to status. But the uniform also fixes the shape of idiosyncratic judgment, and the hierarchy a given α produces can depend on that shape as well as on its spread. Here I repeat the analysis with $\epsilon \sim \text{Beta}(a, a)$ for $a = 1, 2, 5, 7$. Each distribution is symmetric on $[0, 1]$, and the uniform is the $a = 1$ case. I generate all four distributions from the same uniform draws, transformed by each distribution's quantile function (common random numbers). Every evaluator therefore holds the same position in the noise ranking under every distribution; only the distance between draws changes.

Figure A2 reports simulated hierarchy strength across status weights and supply–demand ratios for each noise distribution and three supply distributions. The amplification mechanism holds under all four. When supply equals demand ($r = 1$), status weight has little effect on upward mobility under uniform supply. When candidates outnumber positions, small weights produce steep hierarchies. The distributions differ where the status signal is weak. With uniform supply and $r = 5$, a weight of $\alpha = 0.1$ raises hierarchy strength from about 0.50 under pure noise to 0.86 under the uniform, compared with 0.78 under Beta$(2, 2)$ and 0.72 under Beta$(5, 5)$ and Beta$(7, 7)$. By $\alpha = 0.5$ the four curves are indistinguishable, plateauing at a hierarchy strength of about 0.95.

How much hierarchy a weak status signal produces depends on the spacing among the most favorable draws. Among 1,250 draws, the top 50 span 0.04 under the uniform, 0.11 under Beta$(2, 2)$, and about 0.15 under Beta$(5, 5)$ and Beta$(7, 7)$. Thinner tails spread the luckiest draws apart, so a candidate from a low-status origin who draws well can outrank candidates from high-status origins by a margin that a small status weight cannot close. This spacing rises and then levels off between $a = 5$ and $a = 7$, as the curves do. The uniform specification of the main text is thus the most permissive member of this family, requiring the least status weight to produce a given hierarchy. Under every distribution considered, however, oversupply amplifies small status weights into steep hierarchies, and the differences between distributions vanish as the status signal strengthens.

A.2 Robustness to matching algorithm The matching model in the main text forms matches with candidate-proposing deferred acceptance (Gale and Shapley 1962). This procedure builds in two assumptions. The resulting allocation is stable, so no candidate and department prefer each other to their assigned matches, and among stable allocations it is the one most favorable

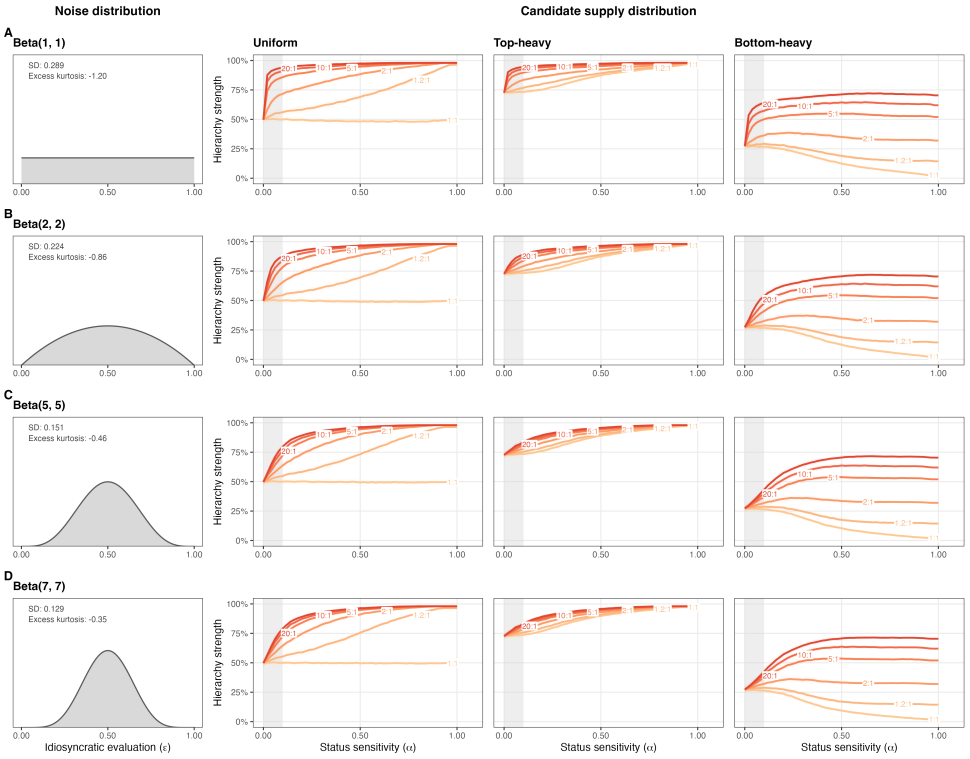


Figure A2: Hierarchy under alternative noise distributions. Rows correspond to $\epsilon \sim \text{Beta}(a, a)$ on $[0, 1]$, unrescaled, for $a = 1, 2, 5, 7$; $a = 1$ is the uniform specification of the main text. The first column shows each noise density with its standard deviation and excess kurtosis. The remaining columns show hierarchy strength, one minus the share of up-hierarchy hires, as a function of the status weight α, by supply–demand ratio, under uniform, top-heavy, and bottom-heavy candidate supply. Lines show means over 100 iterations. The shaded region marks weak status signals ($\alpha \leq 0.1$), as in Fig. 1.

to candidates. The academic job market, however, has no central clearinghouse. Offers are made over time, often with deadlines, and departments screen applicants before choosing among them. Here I repeat the simulations under three alternative procedures that relax these assumptions in turn. Under *department-proposing deferred acceptance*, departments make offers and candidates hold the best offer received. Its outcome is the stable allocation most favorable to departments, so together with the baseline it brackets every stable allocation of a given market. In balanced markets with random preferences, the set of stable allocations (the core of the market) can be large (Pittel 1989). Even the slightest imbalance, however, yields an essentially unique stable allocation (Ashlagi, Kanoria, and Leshno 2017), so whenever candidates outnumber positions ($r > 1$), the choice of which side proposes should matter little. Under *decentralized offers without recall*, departments with open positions make offers in rounds to their highest-ranked candidates not yet hired. A candidate holding several offers accepts the best one, the acceptance is final, and departments whose offers are declined move down their rankings in the next round. The outcome is generally not stable. Under *shortlisting*, each department first screens candidates on a separate evaluation, with the same status weight α and independent noise, and shortlists three candidates per position. Deferred acceptance then proceeds with only shortlisted candidates eligible for each department. Departments whose shortlisted candidates all accept positions elsewhere extend their shortlists by three candidates per open position, and the match is re-solved until every position is filled. As in Appendix A.1, the four procedures are applied to the same draws. Each simulated market is generated once and solved four ways, so differences between procedures reflect how matches form and not which random draws a market received.

Figure A3 reports the share of hires that do not move up the hierarchy under each procedure. Department-proposing deferred acceptance is nearly indistinguishable from the baseline. Whenever candidates outnumber positions, the two procedures place about 98% of candidates in the same department, and their average hierarchies differ by less than 0.1 percentage points. Differences appear only in balanced markets ($r = 1$), where agreement falls to 80% and average hierarchies differ by less than one percentage point. Even there, the shared prestige component narrows the set of stable allocations as the status signal strengthens. When $\alpha \leq 0.05$ and preferences are close to random, the two procedures place only 27% of candidates in the same department, compared with 84% at stronger signals. Which side proposes thus has almost no effect on how much hierarchy a given status weight produces whenever candidates outnumber positions. Dropping stability matters little where

the status signal is weak. For $\alpha \leq 0.2$, decentralized offers produce hierarchies within 2 percentage points of the baseline in every market, even though only two-thirds of candidates are placed in the same department. The two procedures diverge only when the status signal is strong and supply is close to demand, where decentralized offers lower hierarchy by up to 25 percentage points. That region lies outside the regime this paper is concerned with.

Shortlisting slightly weakens amplification at the weakest signals. With uniform supply and a supply–demand ratio of 5, a weight of $\alpha = 0.04$ produces a hierarchy of 0.73 under shortlisting and 0.80 under the baseline, and the gap has nearly closed by $\alpha = 0.2$ (0.88 and 0.89). The differences are confined to weak signals and never change the main result. Under all four procedures, small status weights produce steep hierarchies whenever candidates outnumber positions, and the procedures converge as the status signal strengthens.

A.3 Status, quality, and selection In the main-text model, institutional status is the only signal either side of the market observes, and selection out of academic careers happens only through the match, since candidates who are not hired leave. Here I relax both assumptions. First, departments also observe an individual quality signal that is independent of origin status. Department j evaluates candidate i as

$$U_{ji}^{D} = \alpha P_i + \theta q_i + (1 - \alpha - \theta)\, \epsilon_{ji},$$

where q_i is drawn from a uniform distribution on $[0, 1]$, independently of P_i, and θ is the weight departments place on it, with $\alpha + \theta \leq 1$. Candidate utilities are unchanged. Quality is common to all departments, so it acts as a shared ranking of candidates that is unrelated to their origins and competes directly with the status signal. Second, candidates select into or out of the academic market before matching begins. Each candidate remains in the pool with probability

$$p_i = 0.5 + \gamma\,(P_i - 0.5),$$

truncated to $[0, 1]$. With $\gamma = 0$, half of each origin leaves at random. With $\gamma > 0$, candidates from high-status origins remain more often, and with $\gamma < 0$ they leave more often, as they might in fields where the best-placed doctorates have attractive options outside academia. I simulate a market of 50 institutions with five positions each and uniform supply of five candidates per position before selection, which leaves about 2.5 candidates per position when $\gamma = 0$.

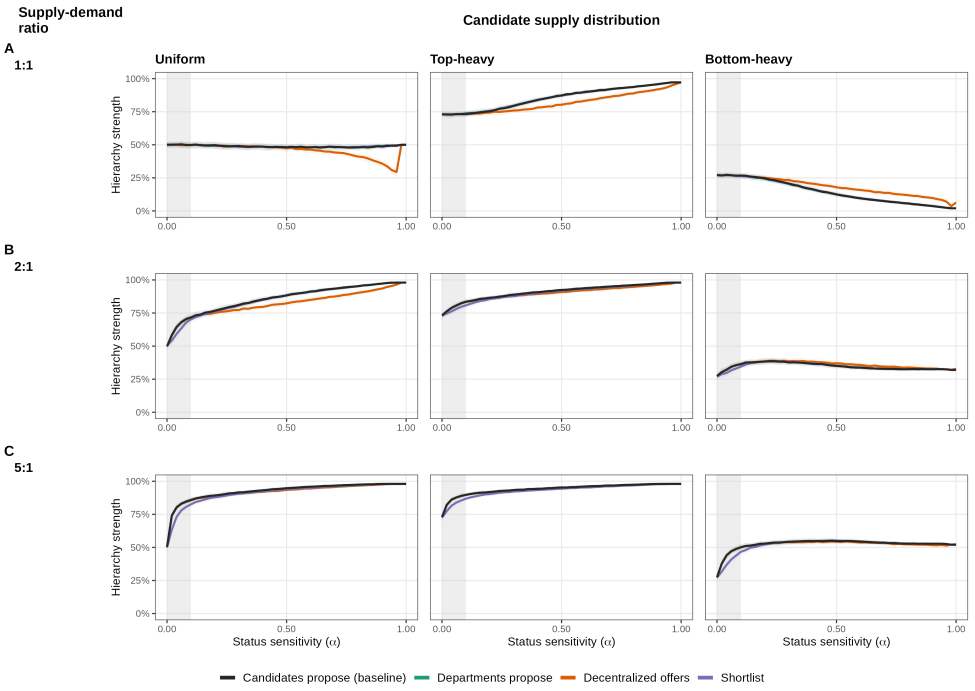


Figure A3: Hierarchy under alternative matching procedures. Rows correspond to supply–demand ratios and columns to candidate supply distributions. Each panel shows the share of hires that do not move up the hierarchy as a function of the status weight α under candidate-proposing deferred acceptance (the main-text baseline), department-proposing deferred acceptance, decentralized offers without recall, and shortlisting of three candidates per position followed by deferred acceptance. All four procedures are applied to the same simulated markets. Lines show means over 100 iterations and bands show the 10th to 90th percentiles. The shaded region marks $\alpha \leq 0.1$, the range of weak status signals defined in Fig. 1.

Figure A4 shows hierarchy strength across selection strength for four quality weights, with the range observed across the 31 calibrated fields shaded (0.77 to 0.92). Selection alone cannot reproduce observed hierarchies. Without a status signal, even the strongest positive selection raises hierarchy only from 0.50 to 0.68. Combined with a weak status signal, however, positive selection is enough. Without quality, a status weight of $\alpha = 0.05$ enters the observed range once γ reaches about 0.4, and $\alpha = 0.1$ does so without any selection. Quality raises the status weight required, and positive selection then does more of the work. At $\theta = 0.1$, a weight of $\alpha = 0.1$ reaches the observed range only with $\gamma \geq 0.6$. At $\theta = 0.2$, the same requires $\alpha = 0.2$, and at $\theta = 0.5$ only $\alpha \geq 0.3$ with strong positive selection does. Negative selection works in the opposite direction, so that fields losing their best-placed candidates to other careers need a stronger status signal to sustain the same hierarchy.

Observed hierarchies are thus consistent with many combinations of the three forces, but not with any of them alone. A weak status signal reproduces them when departments place little weight on quality independent of origin or when candidates from high-status origins remain in the academic market more often. When departments weigh such quality heavily, the observed hierarchy requires either a considerably stronger status signal or strong positive selection.

B. Robustness of the empirical calibration

B.1 Stationarity of the supply distribution The calibration combines sources that refer to different cohorts of faculty. Supply and demand totals are both flows over the same decade. Supply is SED doctorate production in 2011–2020, and demand is the number of new faculty in the hiring network of Wapman et al. (2022), faculty who entered the 2011–2020 sample within three years of their doctorate. The supply–demand ratio thus compares doctorates and entry-level positions over the same period. The hierarchy to which the model is calibrated, however, and the distribution of hires across departments come from the network's full edge list. That list records the stock of faculty employed during the decade, most of whom were hired before 2011 and trained in earlier doctoral cohorts. The null models and the calibration use each institution's share of doctorate production within a field, so this mismatch matters only if those shares have shifted substantially over time. If they have not, production shares in 2011–2020 are a reasonable stand-in for those of the cohorts that make up the faculty stock.

To test whether the production distribution is stationary, I compare institutions' 2011–2020 shares with their shares in the decades in which most

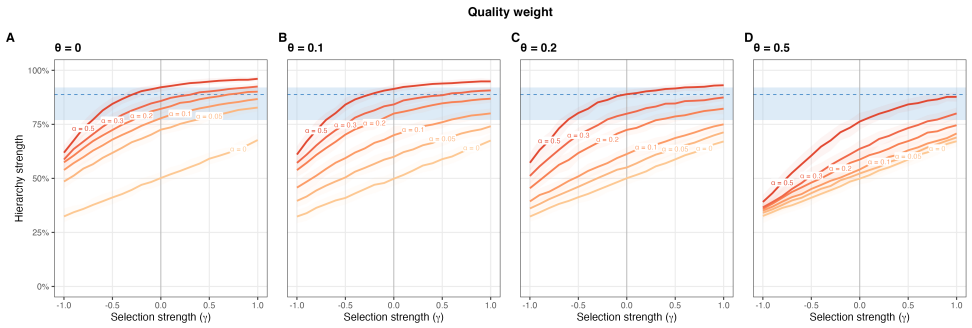


Figure A4: Hierarchy under status, quality, and selection. Panels correspond to the weight departments place on candidate quality independent of origin status, $\theta = 0, 0.1, 0.2, 0.5$. Each panel shows hierarchy strength as a function of selection strength γ, the gradient in the probability that a candidate remains in the academic market by origin status, for status weights α from 0 to 0.5. Lines show means over 100 iterations and bands show the 10th to 90th percentiles. The shaded horizontal band marks the range of hierarchy strength observed across the 31 calibrated fields, and the dashed line their median. Supply is uniform, with five candidates per position before selection.

of the faculty stock was trained. The SED records doctorate production by institution and field from 1958 onward, and the same institutions and fields can be followed across the whole period. Institution names and identifiers are constant across years, all 62 SED fields appear in every decade, and institutions matched to the hiring network account for 92–94% of doctorates in each decade. For each of the 31 fields, I compute each matched institution's share of doctorate production in 1981–1990, 1991–2000 and 2001–2010 and correlate it with its share in 2011–2020 (Figure B1).

Adjacent decades are about as stable as the two halves of 2011–2020, with median correlations of 0.94 and 0.95. Over longer spans shares drift gradually. The median correlation with 2011–2020 is 0.95 for 2001–2010, 0.89 for 1991–2000 and 0.83 for 1981–1990. Pooling 1981–2010, the period in which most of the faculty stock was trained, gives a median correlation of 0.92 with 2011–2020.

Education administration is the one exception, with correlations between 0.17 and 0.30 against 2011–2020. These reflect abrupt changes in the recorded series rather than gradual drift. The field's largest producer before 2005 records no doctorates in it afterward, and recorded production across all in-

stitutions falls by more than half between 2009 and 2011. Excluding this field, the lowest correlation between the pooled 1981–2010 shares and 2011–2020 is 0.81. The distribution of doctorate production is thus stable over the period in which the faculty stock was trained, and 2011–2020 production shares describe the supply behind the observed hierarchy almost as well as the supply of new entrants.

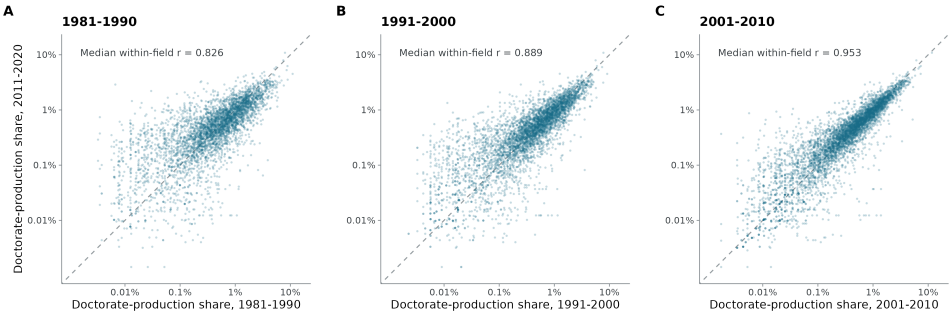


Figure B1: Long-run stability of doctorate production. Each point is an institution–field pair in the 31 analyzed fields, comparing its share of SED doctorate production among matched hiring institutions in 1981–1990 (A), 1991–2000 (B) and 2001–2010 (C) with its share in 2011–2020. Both axes are logarithmic, and the dashed line marks equality. Pairs with zero production in one of the two decades are omitted from the plot but included in the correlations.

B.2 Robustness to academic field aggregation The SED and the hiring network of Wapman et al. (2022) classify doctorates and faculty at different levels of detail. The SED reports doctorate production for broad fields, while the hiring network distinguishes 107 narrower subfields. To pair supply with demand, I assign each subfield to a broad SED field, producing the 31 fields analyzed in the main text. In 14 of them the broad field and the subfield coincide, as in chemistry, history and sociology. The remaining 17 combine between 2 and 13 subfields, as in biological sciences and health sciences, which each combine 13.

The fields that consist of a single subfield provide a benchmark that does not depend on how the two classifications are reconciled, and they support the main result (Figure B2). Their median calibrated status weight is 0.016, with an interquartile range of 0.007 to 0.028, and all 14 fall below 0.1. The

largest is 0.048, in political science. The two estimates above 0.1 among fields that combine several subfields, other engineering and business management, are larger than any in the single-subfield fields.

Combining subfields does, however, affect the calibration targets. The hierarchy of each field is measured against a single ranking of institutions estimated on its hiring network, and when several subfields are pooled, that ranking fits each of them less closely than its own would. More hires then count as moves up the hierarchy. In all 17 fields that combine subfields, the pooled target is lower than the hire-weighted average of the subfields' own hierarchies, by a median of 1.8 percentage points and at most 6.3 points, in health sciences. These fields have a median calibrated weight of 0.017, compared with 0.016 in the single-subfield fields. The single-subfield fields therefore give the cleaner estimates, and they are consistent with the overall median of 0.016. The conclusion that weak status signals are enough to reproduce observed hierarchies does not depend on how subfields are grouped.

B.3 Selection into faculty careers and subsequent sorting The baseline model starts from doctorate production and lets the match determine both who obtains a faculty position and where they are placed. Its calibrated status weight therefore combines two processes, selection into faculty careers and sorting across departments. Here I separate them.

In the first stage, I describe selection with a single status gradient. Let S_{if} denote doctorate production at institution i in field f, and $P_{if} \in [0, 1]$ its status rank. The predicted share of faculty trained at i is

$$\pi_{if}(\gamma_f) = \frac{S_{if} \exp(\gamma_f P_{if})}{\sum_k S_{kf} \exp(\gamma_f P_{kf})}, \tag{1}$$

and I choose γ_f to match the observed faculty-origin shares as closely as possible. The model is descriptive and does not distinguish participation in the academic market from hiring success or retention. The fitted gradient is positive in all 31 fields, with a median of 2.81 (range 0.32–5.20), so doctorates from high-status institutions are over-represented among faculty in every field. Tilting doctorate production toward high-status institutions with a single gradient closes about 38% of the gap between production and observed faculty origins, reducing the median share of faculty that would need to be reassigned from 26% to 16% (Figure B3A).

In the second stage, I take the selected candidates as given and calibrate a sorting weight α_f that reproduces the observed share of upward placements when every selected candidate is hired. Both specifications, with fitted and

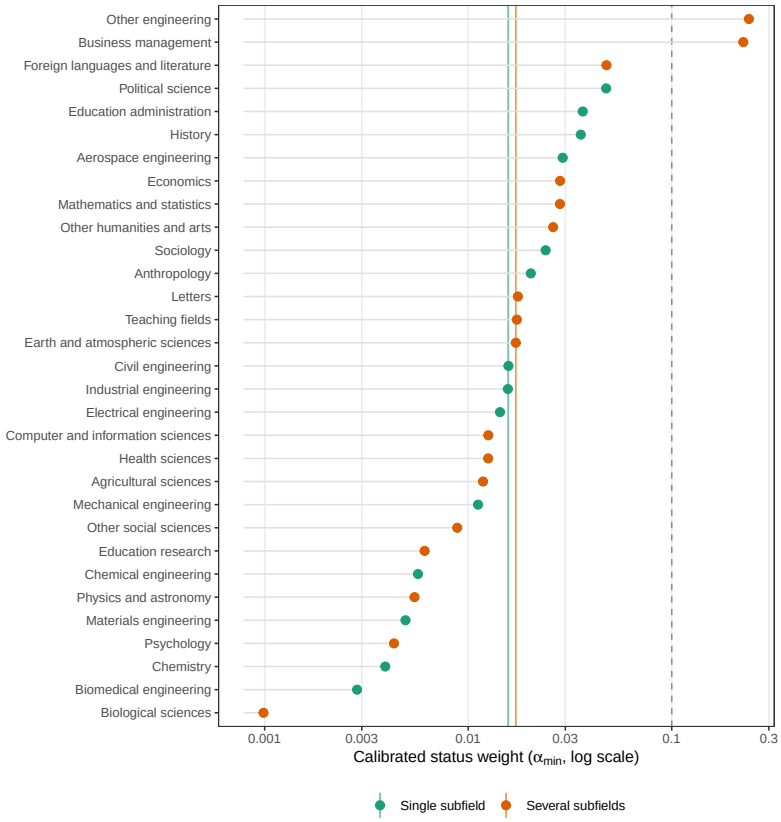


Figure B2: Calibrated status weight by field. Each point is one of the 31 fields analyzed, distinguishing fields that consist of a single subfield of the hiring network from fields that combine several. The horizontal axis is logarithmic. Solid vertical lines mark each group's median, and the dashed line marks $\alpha = 0.1$.

with exactly observed faculty origins, reproduce the target in all 31 fields, with median sorting weights of 0.282 and 0.287 (Figure B3B). Once selection has shaped the pool, a considerably stronger status signal is needed to sort candidates across departments, because the oversupply that amplifies weak signals in the baseline model has already been absorbed by selection.

The same logic explains how the baseline estimate depends on the assumed candidate pool. I vary the fraction q of doctorates who enter the applicant market, holding their origin shares fixed, so that the ratio of applicants to positions becomes $q \sum_i S_{if}/D_f$, where D_f is new-faculty demand. Across the 29 fields that can fill their positions at every participation rate, the median calibrated weight rises from 0.016 at full participation to 0.042 at half participation and 0.220 at one-quarter participation (Figure B3C). These calibrations use the same fixed field-specific rankings, institutional demand allocation, 1,000 vacancies, and fitting routine as the main estimates. The full-participation points are the corresponding main-text fits, restricted here to the 29 fields feasible at every participation rate. The fewer doctorates compete for positions, the stronger the status signal required.

The observed hierarchy can therefore also be reproduced by separating selection into faculty careers from subsequent sorting. The baseline weight is best read as the status signal required when selection and sorting operate through a single match on the full pool of doctorates, and not as a measure of sorting among those who have already been selected.

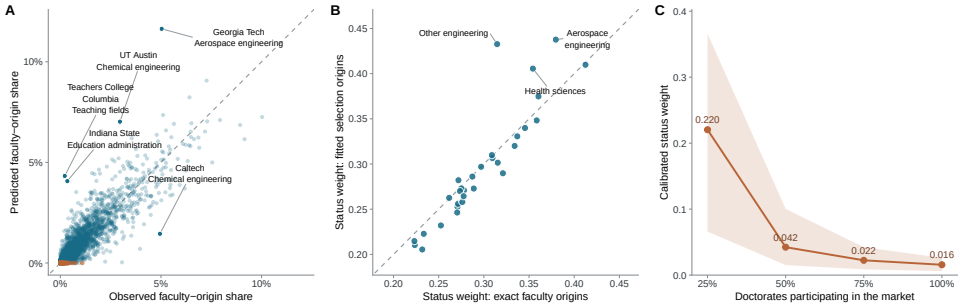


Figure B3: Selection, sorting and the candidate pool. (A) Observed against predicted faculty-origin shares from the fitted selection model across 31 fields. Orange points indicate zero observed doctorate production. (B) Calibrated sorting weights using exact faculty origins against fitted selection origins. Each point is a field, and dashed lines indicate equality. (C) Calibrated status weight by assumed doctorate participation, with 1,000 simulated vacancies. The line and band show the median and interquartile range across the 29 fields feasible at all participation rates.

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